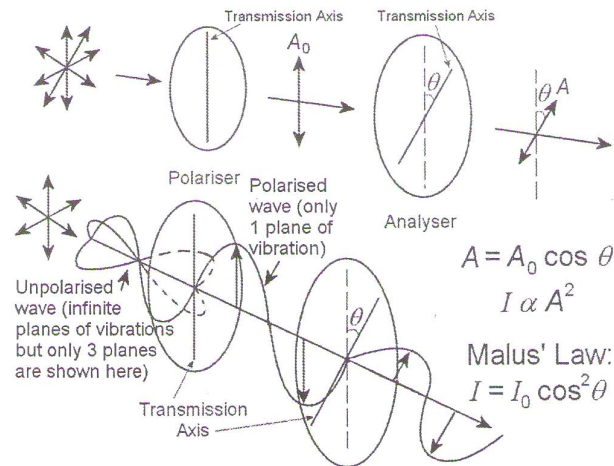


- 7 (a) A progressive wave is a wave in which the waveform advances and there is a transfer of energy along the direction of propagation of the wave. A transverse wave is a wave in which the vibrations are perpendicular to direction of propagation of the wave.

Note: A clear reference to direction of energy of energy transfer is required.

- (b) In a polarised wave, the vibrations are in only one plane; whereas in an unpolarised wave there are infinite planes of vibrations.



- (c) (i) amplitude = A

- (ii) intensity = I

- (iii) $I = kA^2$, where k is the constant of proportionality.

- (d)

Angle	Amplitude	Intensity
180°	A	I
90°	0	0
60°	$0.5 A$	$0.25 I$

- (e) (i) The principle of superposition states that when 2 or more waves of the same kind exist simultaneously at a point in space, the displacement of the resultant wave at that point is the vector sum of displacements of the individual waves.

- (ii) When the sound waves reach open end, it is reflected due to changes in boundary conditions.

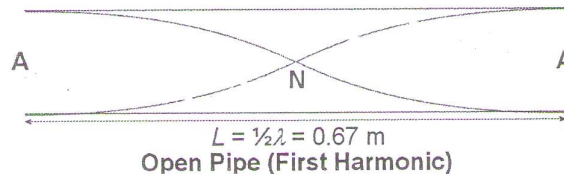
Since boundary is an open end, the reflected wave experiences no phase change at the point of reflection and interferes constructively with the incident wave to produce an antinode there with twice the amplitude.

The incident and reflected waves also undergo superposition throughout the tube to produce regions of high and low amplitude along the pipe.

The highest amplitude occurs during constructive interference giving rise to antinodes. While the zero amplitudes occur when destructive interference happens when the waves meet in anti-phase.

Note: Must mention reflection due to boundary conditions at open end!

- (f) (i) 1. The displacement antinodes are marked as A.
2. The displacement nodes are marked as N.



- (ii) The distance between adjacent antinodes is half a wavelength

$$\frac{1}{2} \lambda = 0.67 \text{ m}$$

$$\lambda = 1.34 \text{ m}$$

$$v = f \lambda = 335 \text{ ms}^{-1}$$

- (g) The underestimation is due to end correction of pipes whereby antinodes are not exactly at the ends of the pipe but located slightly outside. The nodes however remain at the same position as shown.